

KAMALESHWARAN P

Mechanical Engineer · Design Specialist · FEA & Simulation · AI-Integrated Systems

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Coimbatore, Tamil Nadu

PERSONAL INFO

DOB: 17 February 2007

Blood Group: B+ (ve)

Nationality: Indian

Location: Coimbatore, Tamil Nadu

Languages: English (Cambridge B2 Upper-Intermediate), Tamil (Native)

EDUCATION

B.E. Mechanical Engineering

Sri Ramakrishna Institute of Technology
CGPA: 8.54/10 | 2nd Year (Ongoing) | Expected 2028

12th HSC — 80.1%

Kongunadu Matric Hr. Sec. School, Namakkal, 2023

10th SSLC — 87.6%

Kongunadu Matric Hr. Sec. School, Namakkal, 2021

TECHNICAL SKILLS

CAD / Design

AutoCAD · SolidWorks · Fusion 360 · Solid Edge

Simulation / FEA

ANSYS Workbench · Fluent · Thermal · CFD

Engineering Tools

MATLAB · Simulink · GD&T · 3D Printing

Embedded Systems

Arduino · ESP32/ESP8266 · UART/I2C/SPI

AI / Software

OpenCV · TensorFlow · SLAM · Neural Networks

Programming

C · C++ · Embedded C · HTML5 · CSS3 · JavaScript

Dev Tools

Git/GitHub · VS Code · Arduino IDE · Canva

LEADERSHIP

Design Head

ASME SRIT Student Section, 2023–Present

Kart Team Lead & Design Head

SRIT Kart Team — TN Karting S02, Title Won

Analysis Head

Kart Team — ANSYS FEA & Structural Analysis

Lead Member

SAE India, SRIT Chapter, 2023–Present

CAREER OBJECTIVE

Mechanical Engineering student at Sri Ramakrishna Institute of Technology (CGPA: 8.54/10) with advanced proficiency in CAD, FEA, and product development. Passionate about design engineering and building intelligent systems at the intersection of mechanical precision and technology. Cross-disciplinary skills in embedded systems, AI, and web development make me a versatile, future-ready engineer. Seeking a senior-level role in design, analysis, or product development, with openness to contribute wherever impact is greatest.

INTERNSHIP EXPERIENCE

Mechanical Engineering Intern — Sri Jayasurya Engineering, Perur, Coimbatore 15–29 May 2026

- Contributed to robotic arm design/assembly — kinematic linkage design and fabrication workflows.
- Performed quality inspection: dimensional tolerancing, component verification and instrumentation.

Web Development Intern — ZyvOne Technologies, Udumalpettai, Tiruppur Jun 2026 (2 Wks)

- Engineered a hospital management website with appointment scheduling and responsive patient UI.
- Developed a jhumka-jewellery e-commerce platform with catalogue and mobile-responsive design.

Virtual Site Development Intern — Sri Ramakrishna Institute of Technology 2024 (3rd Sem Gap)

- Designed a virtual campus platform with interactive mapping and multimedia integration.

KEY PROJECTS

Kart Design & Analysis — TN Karting S02 | Team Lead + Design + Analysis Head — Title Won

- Led 3D CAD modelling, chassis geometry & suspension layout to full regulatory compliance.
- Conducted ANSYS FEA on chassis & critical components, driving the team to a title win.
- Led team coordination, fabrication oversight & competition preparation end-to-end.

Autonomous Rover Navigation — Caterpillar Autonomy Challenge, Shastra IIT Madras Jan–Mar 2026

- Engineered SLAM & monocular-vision navigation for real-time obstacle detection — Semi-Finalist (National).
- Delivered full system integration across mechanical frame, embedded control & navigation software.
- Collaborated cross-functionally on structural analysis, motor selection & hardware-software co-design.

AI-Powered Laser QR Marking & Monitoring — Indian Railways, SIH 2025 Aug 2025–Present

- Developing a ruggedised laser engraving system marking QR/Datamatrix codes for rail-fitting traceability.
- Built AI defect-detection (OpenCV/TensorFlow) and a web app syncing with UDM & TMS platforms.

AI for Assembly — TN-IMPACT 2026, Dassault Systèmes & TN CoE Advanced Mfg.

- Built an AI-driven assembly-line intelligence solution with Krishnakanth J & Nirupathunga M, mentored by Sanjana N & Immanuel R.
- Gained deep exposure to Industry 4.0 thinking and smart manufacturing architecture.

PointIQ — Real-Time Sports Scoreboard System Independent Development

- Designed a real-time scoreboard and score-management system for live sporting events (Arduino, ESP32).
- Implemented multi-module UART/I2C/SPI communication for reliable sensor & component interfacing.

Heat Exchanger CFD Studies — Pebble Bed & Shell-Tube with Parabolic Solar Collector

- Ran ANSYS Fluent CFD on pebble-bed heat transfer, pressure drop & thermal storage efficiency (Dr. N.G.P Institute).
- Evaluated shell-&-tube exchanger performance with parabolic solar collectors, published at SRM Institute of Technology.

CERTIFICATIONS

- ◆ MATLAB Onramp — MathWorks
- ◆ Simulink Onramp — MathWorks
- ◆ Innovations in Automation & Machine Vision
- ◆ Neural Networks & AI Model Creation
- ◆ Innovating with GraphRAG
- ◆ A Path to a Welding Engineer as Career
- ◆ How to Manage Events Like a Pro
- ◆ Effective Sales & Marketing Strategies
- ◆ Learnathon — SRIT

MEMBERSHIPS

- ASME — Design Head, SRIT Chapter
- SAE India — Lead Member, SRIT Chapter
- Rotaract Intl. — Int'l Service Director, SRIT
- IIC / HIVE — Member & Innovation Ambassador
- ASPOMERS — Active Member, SRIT

PATENTS (3× CO-INVENTOR, INDIA 2026)

Patent 01 — Published — India, 10-04-2026 | IN202641039458 A1

Augmented Reality Based Intelligent Assembly Guidance & Torque Monitoring System for Industrial Valve Manufacturing

Co-developed an AR-based system providing real-time assembly guidance and torque monitoring for industrial valve manufacturing.

Patent 02 — Published — India, 27-03-2026 | IN202641034420 A1

Extended Reality Teleoperation System with 360° Video Streaming & Real-Time Digital Twin Monitoring for Remote Environment Exploration

Co-developed an XR teleoperation system for a mobile rover, integrating 360° streaming, LiDAR-based digital twin monitoring & VR control.

Patent 03 — Published — India, 19-06-2026 | IN202641057635 A1

AI-Based Automatic Visual Inspection System for Small Scale MSME Manufacturing

Co-invented an AI-powered inspection system using dual robotic camera arms and deep-learning defect detection for MSME manufacturing.

RESEARCH PAPERS

- "Performance Evaluation of Shell & Tube Heat Exchanger with Parabolic Solar Collector" — SRM Institute of Technology, National Conference (Published, Peer-Reviewed).
- "Energy Storage & CFD Analysis of Pebble Bed Heat Exchanger" — Dr. N.G.P Institute of Technology, National Conference.

AWARDS & ACHIEVEMENTS

- 3× Patent Co-Inventor — India, 2026 (AR Assembly Guidance, XR Teleoperation, AI Quality Inspection).
- Tamil Nadu Karting Championship — Title Winner — as Kart Team Lead, Design Head & Analysis Head.
- Smart India Hackathon — 1st Place (Internal Round), advancing to the national stage.
- National-Level Finalist — Caterpillar Autonomy Challenge (Semi-Finalist), MSME Idea Hackathon, Statathon & Pivot Challenge.
- National & University Research Presenter — 2 published conference papers plus FORSCH'2025, Anna University.
- Two-Time Open State Weightlifting Champion (60 kg Category).

PROFESSIONAL MEMBERSHIPS

Organization	Role	Duration
ASME	Design Head, SRIT Chapter	2023–Present
SAE India	Lead Member, SRIT Chapter	2023–Present
Rotaract Intl.	Int'l Service Director, SRIT	2023–Present
IIC / HIVE	Member & Innovation Ambassador	2023–Present
ASPOMERS	Active Member, SRIT	2023–Present

REFERENCES

Mr. Immanuel R
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Ms. Sanjana N
Professor, Dept. of Computer Science & Engg.
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All information accurate as of June 2026 | [linkedin.com/in/kamaleshwaran-p](#) | [github.com/Kamaleshwaranbadboy](#)